

Long-range order and spontaneous symmetry breaking in quantum spin systems and hard-core bosons on a lattice

part 2 LRO and SSB in the equilibrium states of the classical Ising model

***Advanced Topics in
Statistical Physics
by Hal Tasaki***

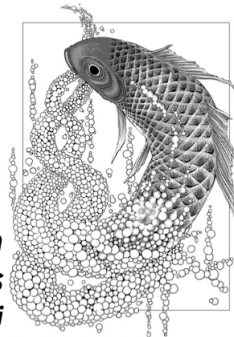


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§ model and basic symmetry

Hamiltonian of the ferromagnetic Ising model

$$\hat{H}_L^0 = - \sum_{\{u,v\} \in \mathcal{B}_L} \hat{S}_u^{(z)} \hat{S}_v^{(z)} \quad (1)$$

symmetry $\hat{U} = \hat{U}_L^{(x,\pi)} \quad (2)$

$$\hat{U}^\dagger \hat{S}_u^{(\alpha)} \hat{U} = \begin{cases} \hat{S}_u^{(\alpha)} & , \alpha = x \\ -\hat{S}_u^{(\alpha)} & , \alpha = y, z \end{cases} \quad (3)$$
$$\hat{U}^\dagger \hat{H}_L^0 \hat{U} = \hat{H}_L^0 \quad (4) \text{ invariant under } \hat{U}$$

order operator $\hat{O}_L^{(z)} = \sum_u \hat{S}_u^{(z)} \quad (5)$

$$\hat{U}^\dagger \hat{O}_L^{(z)} \hat{U} = -\hat{O}_L^{(z)} \quad (6)$$

$$\langle \hat{O}_L^{(z)} \rangle = 0 \quad (7) \text{ if } \langle \dots \rangle \text{ is } \hat{U}\text{-invariant}$$

$$\text{(i.e. } \langle \hat{U}^\dagger \hat{A} \hat{U} \rangle = \langle \hat{A} \rangle \text{ for any } \hat{A} \quad (8) \text{)}$$

§ equilibrium states at low temperatures

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equilibrium state

$$\langle \dots \rangle_L^{\beta,0} = \frac{\text{Tr}[(\dots) e^{-\beta \hat{H}_L^0}]}{\text{Tr}[e^{-\beta \hat{H}_L^0}]} \quad (1)$$

$$\text{Tr}[\hat{A}] = \sum_{\phi \in \mathcal{H}} \langle \phi | \hat{A} | \phi \rangle \quad (2)$$

theorem: let $d \geq 2$. for sufficiently large $\beta > 0$, there is $q_0(\beta) > 0$ s.t.

$$\langle \hat{S}_u^{(z)} \hat{S}_v^{(z)} \rangle_L^{\beta,0} \geq q_0(\beta) \quad (3) \quad \text{for any } u, v \in \Lambda_L \text{ and any } L$$

and hence

$$\left\langle \left(\frac{\hat{O}_L^{(z)}}{L^d} \right)^2 \right\rangle_L^{\beta,0} = \frac{1}{L^{2d}} \sum_{u,v \in \Lambda_L} \langle \hat{S}_u^{(z)} \hat{S}_v^{(z)} \rangle_L^{\beta,0} \geq q_0(\beta) \quad (4)$$

long-range order (LRO) detected by the order operator $\hat{O}_L^{(z)}$

on the other hand, the $\hat{U}$ -symmetry shows $\langle \hat{O}_L^{(z)} \rangle_L^{\beta,0} = 0 \quad (5)$

no spontaneous symmetry breaking (SSB)

LRO + no SSB implies

fluctuation of $\hat{O}_L^{(2)}/L^d$

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$$\sqrt{\left\langle \left(\frac{\hat{O}_L^{(2)}}{L^d} \right)^2 \right\rangle_L - \left[\left\langle \frac{\hat{O}_L^{(2)}}{L^d} \right\rangle_L \right]^2} \geq \sqrt{q_0} > 0 \quad (1)$$

The density of a macroscopic quantity has nonzero fluctuation
no stable macroscopic physics!

UNPHYSICAL!!

physical state should have LRO + SSB

Griffiths theorem

can be obtained in
several manners

- infinitesimal symmetry breaking field
- symmetry breaking boundary conditions
- pure state decomposition

§ equilibrium state under infinitesimal symmetry breaking field

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Hamiltonian with uniform external field h

$$\hat{H}_L^h = - \sum_{\{u,v\} \in \mathcal{B}_L} \hat{S}_u^{(z)} \hat{S}_v^{(z)} - h \sum_{u \in \Lambda_L} \hat{S}_u^{(z)} \quad (1)$$

$\hat{S}_u^{(z)} = \hat{O}_L^{(z)}$

$$\langle \dots \rangle_L^{\beta, h} = \frac{\text{Tr}[(\dots) e^{-\beta \hat{H}_L^h}]}{\text{Tr}[e^{-\beta \hat{H}_L^h}]} \quad (2)$$

for $d \geq 2$ and sufficiently large $\beta > 0$

spontaneous symmetry breaking!

$$\lim_{h \downarrow 0} \lim_{L \uparrow \infty} \left\langle \frac{\hat{O}_L^{(z)}}{L^d} \right\rangle_L^{\beta, h} = m^*(\beta) \geq \sqrt{q_0(\beta)} > 0 \quad (3)$$

fluctuation of $\hat{O}_L^{(z)}/L^d$ is small for large L

$$\lim_{h \downarrow 0} \lim_{L \uparrow \infty} \sqrt{\left\langle \left(\frac{\hat{O}_L^{(z)}}{L^d} \right)^2 \right\rangle_L^{\beta, h} - \left(\left\langle \frac{\hat{O}_L^{(z)}}{L^d} \right\rangle_L^{\beta, h} \right)^2} = 0 \quad (4)$$

PHYSICAL!

remark: pure state decomposition of infinite volume states

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$\hat{A}$ any polynomial of $\hat{S}_u^{(2)}$ with $u \in \mathbb{Z}^d$

infinite volume equilibrium states

$$\langle \hat{A} \rangle^{\beta, 0} = \lim_{L \uparrow \infty} \langle \hat{A} \rangle_L^{\beta, 0} \quad (1)$$

$$\langle \hat{A} \rangle^{\beta, \pm} = \lim_{\hbar \rightarrow \pm 0} \lim_{L \uparrow \infty} \langle \hat{A} \rangle_L^{\beta, \hbar} \quad (2)$$

for $d \geq 2$ and sufficiently large $\beta > 0$

pure state decomposition (mathematically well-defined notion)

$$\langle \dots \rangle^{\beta, 0} = \frac{1}{2} \{ \langle \dots \rangle^{\beta, +} + \langle \dots \rangle^{\beta, -} \} \quad (3)$$

unphysical equilibrium state
with LRO but no SSB

physical equilibrium states
with LRO and full SSB